Design-actionable cost–resilience thresholds for islanded battery–hydrogen microgrids: component drivers, diesel-support robustness, and dynamic feasibility

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Abstract

Islanded and remote microgrids increasingly weigh battery–hydrogen storage against diesel-backed batteries, yet the conditions under which hydrogen is actually justified remain imprecisely defined. The cost, resilience, and frequency-stability dimensions of this choice have each been studied, but usually in separate analyses and for a single optimised design per technology—leaving open which design levers move the economic threshold, whether the resilience advantage is intrinsic to hydrogen or a by-product of its sizing, and how robust the verdict is to the assumed level of diesel support. Using an 8760-h cost-optimal dispatch model driven by real NASA POWER weather for a representative Philippine off-grid island, we bring these dimensions together into a single, design-actionable, multi-axis comparison of a battery–hydrogen portfolio against a PV–diesel–battery portfolio. We find an economic crossover at a delivered diesel cost of 378 USD/MWh. A component-level decomposition shows the frontier is governed primarily by photovoltaic capital cost—because the hydrogen portfolio is intrinsically PV-intensive—rather than by electrolyser or fuel-cell efficiency. A matched-backbone comparison demonstrates that the resilience advantage is structural: on an identical PV–battery spine, the diesel–battery system cannot bridge the worst-case 48-hour outage—it sheds critical load and provides no guaranteed uninterrupted service (worst-case critical-load survivability 0.98)—whereas the battery–hydrogen system sustains the full 48 hours, owing to long-duration hydrogen storage. A graded diesel-support stress test shows that any derating below roughly 50%, any fuel budget below about

12 hours, or any delivery delay collapses the diesel–battery worst-case survivability, and a reduced-order frequency screen confirms that this weakness extends to the dynamic layer during the diesel-trip grid-forming handover. Battery–hydrogen is therefore defensible whenever delivered diesel is costly or diesel support is degradable—precisely the conditions that characterise remote-island contingencies.

Keywords: islanded microgrid, green hydrogen, long-duration storage, resilience, cost threshold, grid-forming inverter, frequency stability

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1. Introduction

Remote and islanded power systems serve a large and growing share of the world's un- and under-electrified population and remain overwhelmingly dependent on diesel generation [1, 2]. In archipelagic countries such as the Philippines, hundreds of small island grids rely on imported diesel whose delivered cost is inflated by maritime logistics and is highly volatile, while the same islands face frequent typhoon-driven disruptions that can sever fuel resupply for days [3, 4, 5]. These systems therefore sit at the intersection of two pressures that are usually studied separately: decarbonisation, which favours displacing diesel with solar and storage, and resilience, which requires riding through multi-day contingencies when external supply fails.

Battery–hydrogen hybrids are increasingly proposed to address both pressures, pairing short-duration batteries with a long-duration hydrogen store—electrolyser, tank, and fuel cell—that can in principle sustain critical load through extended outages without fuel deliveries [6, 7, 8]. Hydrogen subsystems are, however, capital-intensive and round-trip inefficient, so whether they are justified over a conventional PV–diesel–battery design depends on site-specific economics and on how severe and how diesel-dependent the resilience requirement is. Establishing the *thresholds* that separate the two designs—and, crucially, identifying which design levers move those thresholds—is what makes such an analysis actionable for planners rather than merely descriptive.

A substantial literature optimises the sizing and dispatch of hybrid renewable–storage microgrids [9, 10, 11], and a growing body of work examines hydrogen specifically in island and off-grid contexts [12, 13, 14]. Resilience-oriented studies introduce metrics such as expected energy not served and critical-load survivability and couple them to outage scenarios

[15, 16, 17, 18]. In parallel, the power-electronics community has established that inverter-dominated islanded grids require explicit grid-forming sources and that frequency stability—rate of change of frequency (RoCoF) and frequency nadir—becomes a binding design concern as synchronous inertia disappears [19, 20, 21, 22].

Each of these dimensions has been studied, often in depth. What remains uncommon is to bring them together into a single, decision-actionable comparison. In particular: (i) economic thresholds are usually reported for one optimised design rather than attributed to the specific component costs that move them; (ii) competing portfolios are compared as separately optimised designs, which conflates the storage technology with the sizing it induces—controlled comparisons on a common PV–battery backbone, which would isolate the technology's own contribution, are rare; (iii) the level of diesel support during an outage is fixed at one or a few scenarios rather than swept as a continuous robustness boundary that locates where a diesel–battery system matches hydrogen; and (iv) the economic, resilience, and frequency-stability verdicts are typically established in separate studies, so their mutual consistency for a given design is seldom verified [23, 24, 25, 26, 27, 28, 29].

This paper addresses these gaps with a unified, design-actionable threshold analysis of a battery–hydrogen versus a PV–diesel–battery portfolio, built on an 8760-h cost-optimal dispatch model driven by real NASA POWER weather (calendar year 2025, central Visayas) for a representative Philippine off-grid island, with a calibrated representative load profile. The contributions are: (i) a fine cost–resilience frontier locating the economic crossover and its sensitivity to hydrogen capital cost; (ii) a component-level decomposition identifying which cost and efficiency parameters most move that frontier; (iii) a matched-backbone comparison—the methodological centrepiece—that places both portfolios on a common PV–battery spine to isolate the contribution of the flexibility technology from sizing; (iv) a graded diesel-support stress test that treats the diesel-availability assumption as a continuous robustness boundary across derating, fuel-budget, and delivery-delay cases; and (v) a reduced-order frequency-stability screen with explicit grid-forming source assignment for each portfolio and operating mode, integrated into the same framework so the economic, resilience, and dynamic verdicts are mutually consistent. The findings are not the expected truism that hydrogen wins when diesel is dear, but a set of more specific and partly counter-intuitive results: the economic crossover occurs at a delivered diesel cost of 378 USD/MWh; the frontier is governed primarily by PV capital cost—not hydrogen component cost or efficiency—because the hydrogen portfolio is intrinsically PV-intensive; the resilience advantage is structural, persisting even at a matched backbone, where the diesel–battery system cannot bridge the worst-case 48-hour outage—sheds critical load with zero guaranteed survivable hours—against hydrogen's full 48 hours; and this advantage extends to the frequency-dynamic layer, where the diesel–battery system is weakest precisely during the diesel-trip handover that also drives its energy-resilience weakness. Together these results show that battery–hydrogen is defensible whenever delivered diesel is costly or diesel support is degradable, both of which characterise remote-island contingencies.

The remainder of the paper is organised as follows. Section 2 describes the system model, cost formulation, resilience evaluation, and the reduced-order dynamic screen; Section 3 gives the case study and assumptions. Section 4 presents the threshold frontier and its component drivers, the matched-backbone comparison, the diesel-support robustness analysis, and the frequency-stability screen. Section 5 discusses design implications and limitations, and Section 6 concludes.

2. Methods

We compare three fixed-architecture portfolios for an islanded microgrid: a PV–battery system (battery-only), a PV–diesel–battery system (diesel-battery), and a PV–battery–hydrogen system (battery-hydrogen). For each portfolio we (i) solve an 8760-h cost-optimal dispatch, (ii) accumulate an annualised total cost, (iii) evaluate critical-load resilience over a stochastic outage set, and (iv) screen primary-frequency dynamics with a reduced-order model. Sections 2.5 and 2.6 define the cost–resilience threshold, the matched-backbone construction, and the diesel-support robustness sweep that the Results build on.

2.1. Sets, variables, and parameters

Let $t \in \mathcal{T} = \{1, \ldots, 8760\}$ index the hours of one year, with time step $\Delta t = 1\,\mathrm{h}$. The hourly decision variables are the used PV power p_t^{pv}, battery charge and discharge powers p_t^{ch}, p_t^{dis} and state of charge e_t, diesel power p_t^{dg}, fuel-cell power p_t^{fc}, electrolyser power p_t^{el}, hydrogen inventory m_t, and unserved load u_t (all ≥ 0). Fixed inputs are the demand D_t and the available PV $\hat{p}_t^{\mathrm{pv}}$. Installed sizes are $\bar{S}^{\mathrm{pv}}$, battery power $\bar{S}^{\mathrm{bp}}$ and energy $\bar{S}^{\mathrm{be}}$, $\bar{S}^{\mathrm{dg}}$, $\bar{S}^{\mathrm{fc}}$, $\bar{S}^{\mathrm{el}}$, and tank $\bar{S}^{\mathrm{tk}}$ (kg). Round-trip parameters are battery charge/discharge efficiencies $\eta^{\mathrm{ch}}, \eta^{\mathrm{dis}}$, electrolyser and fuel-cell LHV efficiencies $\eta^{\mathrm{el}}, \eta^{\mathrm{fc}}$, and the hydrogen lower heating value LHV = 33.3 kWh kg^{-1}. Cost parameters are summarised in Table 1.

2.2. Optimal dispatch model

For fixed capacities, the hourly dispatch minimises annual operating cost,

$$\begin{aligned} \min \sum_{t\in\mathcal{T}} \Big[&(c^{\mathrm{fuel}} + c^{\mathrm{co_2}}\varepsilon^{\mathrm{dg}})\, p_t^{\mathrm{dg}} + c^{\mathrm{thr}}\,(p_t^{\mathrm{ch}} + p_t^{\mathrm{dis}}) \\ &+ c^{\mathrm{el}}\, p_t^{\mathrm{el}} + c^{\mathrm{fc}}\, p_t^{\mathrm{fc}} + c^{\mathrm{pen}}\, u_t \Big] \Delta t, \end{aligned} \tag{1}$$

where c^{fuel} is the delivered diesel cost (USD MWh^{-1} of electrical output), $\varepsilon^{\mathrm{dg}} = 0.70$ t CO_2 MWh^{-1} the diesel emission factor, $c^{\mathrm{co_2}}$ the carbon price, $c^{\mathrm{thr}} = 1$ USD MWh^{-1} a battery throughput (cycling) cost charged on both charge and discharge, $c^{\mathrm{el}} = 2$ and $c^{\mathrm{fc}} = 4$ USD MWh^{-1} small electrolyser and fuel-cell variable-O&M terms, and $c^{\mathrm{pen}} = 10{,}000$ USD MWh^{-1} a high unserved-energy penalty that makes load-shedding a last resort within the dispatch (a modelling device, distinct from the value of lost load used for resilience in Section 2.4). With $\Delta t = 1$ h, each hourly power term equals its energy. Subject to, for all t, PV allocation with explicit curtailment p_t^{cu} and the system power balance,

$$p_t^{\mathrm{pv}} + p_t^{\mathrm{cu}} = \hat{p}_t^{\mathrm{pv}} = \bar{S}^{\mathrm{pv}}\,\gamma_t, \tag{2}$$

$$p_t^{\mathrm{pv}} + p_t^{\mathrm{dis}} + p_t^{\mathrm{dg}} + p_t^{\mathrm{fc}} + u_t \;=\; D_t + p_t^{\mathrm{ch}} + p_t^{\mathrm{el}}, \tag{3}$$

where $\gamma_t \in [0,1]$ is the PV capacity factor. Storage dynamics use battery charge/discharge efficiencies $\eta^{\mathrm{ch}} = \eta^{\mathrm{dis}} = 0.95$ and hydrogen conversion coefficients $\kappa^{\mathrm{el}} = 10^3\eta^{\mathrm{el}}/\mathrm{LHV} = 20.1$ kg MWh^{-1} and $\kappa^{\mathrm{fc}} = 10^3/(\eta^{\mathrm{fc}}\,\mathrm{LHV}) = 60.0$ kg MWh^{-1},

$$\begin{aligned} e_t &= e_{t-1} + \eta^{\mathrm{ch}} p_t^{\mathrm{ch}}\Delta t - \tfrac{1}{\eta^{\mathrm{dis}}} p_t^{\mathrm{dis}}\Delta t, \\ m_t &= m_{t-1} + \kappa^{\mathrm{el}} p_t^{\mathrm{el}}\Delta t - \kappa^{\mathrm{fc}} p_t^{\mathrm{fc}}\Delta t, \end{aligned} \tag{4}$$

with bounds $0 \le e_t \le \bar{S}^{\mathrm{be}}$, $0 \le m_t \le \bar{S}^{\mathrm{tk}}$, and rated-power limits $p_t^{\mathrm{ch}}, p_t^{\mathrm{dis}} \le \bar{S}^{\mathrm{bp}}$, $p_t^{\mathrm{dg}} \le \bar{S}^{\mathrm{dg}}$, $p_t^{\mathrm{fc}} \le \bar{S}^{\mathrm{fc}}$, $p_t^{\mathrm{el}} \le \bar{S}^{\mathrm{el}}$. Both stores are initialised at 50% of capacity and constrained to return to that level at year-end, giving an annually self-consistent cycle. The implied hydrogen round-trip efficiency is $\kappa^{\mathrm{el}}/\kappa^{\mathrm{fc}} = 33.5\%$. The program is a linear program solved with HiGHS; capacities enter as fixed bounds, so the diesel cost and carbon price affect dispatch while capital costs do not.

2.3. Cost formulation and levelised cost

The reported annual cost adds annualised capital and fixed O&M to the optimised operating cost. Capital is annualised with the capital recovery factor

$$\mathrm{CRF} = \frac{i\,(1+i)^L}{(1+i)^L - 1}, \qquad i = 0.07,\ L = 20\,\mathrm{yr} \ \Rightarrow\ \mathrm{CRF} = 0.0944. \tag{5}$$

With unit capital costs c_k^{cap} over components k (PV, battery power, battery energy, diesel, fuel cell, electrolyser, and tank), the total overnight capital is $\mathrm{CAPEX} = \sum_k c_k^{\mathrm{cap}} \bar{S}_k$, and the total annual cost is

$$\begin{aligned} \mathrm{AC} = {} & \underbrace{\mathrm{CRF}\cdot\mathrm{CAPEX}}_{\text{annualised CAPEX}} + \underbrace{\phi\cdot\mathrm{CAPEX}}_{\text{fixed O\&M}} + \underbrace{c^{\mathrm{fuel}}E^{\mathrm{dg}}}_{\text{fuel}} \\ & + \underbrace{c^{\mathrm{co_2}}\varepsilon^{\mathrm{dg}}E^{\mathrm{dg}}}_{\text{carbon}} + \underbrace{c^{\mathrm{thr}}E^{\mathrm{thr}} + c^{\mathrm{el}}E^{\mathrm{el}} + c^{\mathrm{fc}}E^{\mathrm{fc}}}_{\text{variable O\&M}} + \underbrace{c^{\mathrm{pen}}E^{\mathrm{uns}}}_{\text{unserved}}, \end{aligned} \tag{6}$$

where $\phi = 0.025$ is the fixed-O&M rate; $E^{\mathrm{dg}} = \sum_t p_t^{\mathrm{dg}}\Delta t$ is annual diesel energy; E^{thr} battery throughput; $E^{\mathrm{el}}, E^{\mathrm{fc}}$ electrolyser and fuel-cell throughput; and $E^{\mathrm{uns}} = \sum_t u_t\Delta t$ unserved energy. The levelised cost of electricity is $\mathrm{LCOE} = \mathrm{AC}\,/\sum_t (D_t - u_t)\Delta t$, i.e. annual cost divided by annual served energy.

2.4. Resilience evaluation

Resilience is assessed on a stochastic set of islanding events. For an annual frequency f^{out} and a fixed duration Δ^{out}, the outage set is

$$\Omega(s) = \left\{\, \omega_j = (\tau_j, \Delta^{\mathrm{out}}) : \tau_j \sim \mathcal{U}\{1, 8760\},\ j = 1, \ldots, f^{\mathrm{out}} \,\right\}, \tag{7}$$

with start hours drawn reproducibly from seed s. During each event the controllable sources serve the critical load $D_t^{\mathrm{crit}} = \alpha D_t$ (critical fraction $\alpha = 0.55$) from the inventory carried into the event, using a priority dispatch (PV $\rightarrow$ battery $\rightarrow$ fuel cell $\rightarrow$ diesel, the last subject to Section 2.6). For a single event ω,

$$\begin{aligned} \mathrm{EENS}_\omega &= \sum_{t\in\omega} \left(D_t^{\mathrm{crit}} - s_t\right)\Delta t, \\ \mathrm{CLSR}_\omega &= \frac{\sum_{t\in\omega} s_t}{\sum_{t\in\omega} D_t^{\mathrm{crit}}}, \end{aligned} \tag{8}$$

where s_t is the critical load served in hour t. The survivable duration Σ_ω is the number of contiguous hours from the event start before the first unserved hour. We report the worst case over the event set ($\mathrm{CLSR}_{\min}$, $\mathrm{EENS}_{\max}$, $\Sigma_{\min}$) and—to monetise resilience—a value of lost load VOLL = 5000 USD MWh^{-1} applied to expected EENS, giving the resilience-adjusted cost $\mathrm{AC}^{\mathrm{res}} = \mathrm{AC} + \mathrm{VOLL} \cdot \mathbb{E}_\Omega[\mathrm{EENS}]$. The base case uses $f^{\mathrm{out}} = 4\,\mathrm{yr}^{-1}$, $\Delta^{\mathrm{out}} = 48\,\mathrm{h}$; worst-case statistics are taken over ten seeds unless stated.

2.5. *Cost–resilience threshold and matched backbone*

The economic comparison is summarised by the cost gap $\Delta C(c^{\mathrm{fuel}}, \mu) = \mathrm{AC}_{\mathrm{BH}}(\mu) - \mathrm{AC}_{\mathrm{DB}}(c^{\mathrm{fuel}})$, where μ scales the hydrogen-subsystem capital costs (electrolyser, fuel cell, tank). The *economic crossover* $c^\star(\mu)$ solves $\Delta C(c^\star, \mu) = 0$; the resilience-adjusted crossover uses $\Delta C^{\mathrm{res}} = \Delta C + \mathrm{VOLL}\,(\mathbb{E}[\mathrm{EENS}_{\mathrm{BH}}] - \mathbb{E}[\mathrm{EENS}_{\mathrm{DB}}])$. To separate the storage technology from the capacity it induces, the *matched-backbone* comparison assigns both portfolios a common PV size and battery energy (battery power at an energy-to-power ratio of 6 h), differing only in the flexibility subsystem. Comparing the two on an identical backbone isolates the contribution of the flexibility technology to cost and resilience.

2.6. *Diesel-support robustness*

The base resilience case assumes diesel is unavailable during islanding. To test the sensitivity of the verdict to this assumption, the diesel contribution within an event is parameterised by a derating fraction $\delta \in (0, 1]$, a fuel budget B (hours at rated power), and a start delay h_{d}, giving an effective deliverable power $\delta\,\bar{S}^{\mathrm{dg}}$ available only for $t \geq \tau + h_{\mathrm{d}}$ and only while cumulative diesel energy is below $B\,\bar{S}^{\mathrm{dg}}$. The cases are: D0 unavailable; D1 $\delta \in \{0.25, 0.5, 0.75\}$; D2 $B \in \{3, 6, 12, 24\}\,\mathrm{h}$; D3 $h_{\mathrm{d}} \in \{6, 12, 24\}\,\mathrm{h}$; and D4 a delivered-fuel-cost shock on the annual economics. The hydrogen portfolio carries no diesel and is the invariant reference.

2.7. *Reduced-order frequency-stability screen*

To confirm dynamic feasibility, primary frequency response is screened with an aggregate swing equation and first-order droop on each online source. With deviation Δf from the nominal $f_0 = 50\,\mathrm{Hz}$ and source-power deviations

δP_i,

$$\frac{\mathrm{d}\Delta f}{\mathrm{d}t} = \frac{f_0}{2\,H_{\mathrm{eq}}\,S_{\mathrm{sys}}}\Big[\sum_i \delta P_i - \Delta P_{\mathrm{dist}}(t) - D_{\mathrm{eq}}\Delta f\Big],$$
$$\tau_i \frac{\mathrm{d}\delta P_i}{\mathrm{d}t} = -\delta P_i - K_i\,\Delta f, \tag{9}$$

with system base $S_{\mathrm{sys}} = 1.65\,\mathrm{MW}$ (peak load) and equivalent inertia $H_{\mathrm{eq}} = H_{\mathrm{gf}} S_{\mathrm{gf}}/S_{\mathrm{sys}}$ contributed by the grid-forming source. Each portfolio is assigned an explicit grid-forming source—battery virtual synchronous machine for battery-only and battery-hydrogen, diesel synchronous generator in normal operation for diesel-battery with battery hand-over during a diesel trip—reflecting that an islanded system's frequency reference is a grid-forming source rather than an external grid [19, 20]. Disturbances are a 10% load step, a 50% PV ramp, and a diesel-trip hand-over. Metrics are the frequency nadir, the instantaneous and 500-ms RoCoF, the primary droop offset $\Delta f(\infty)$, and the settling time relative to $\Delta f(\infty)$; screening thresholds are nadir ≥ 49.0 Hz and $|\mathrm{RoCoF}| < 2.0\,\mathrm{Hz\,s^{-1}}$ (islanded; a $1.0\,\mathrm{Hz\,s^{-1}}$ mainland value is flagged for reference) [21, 22]. The model represents primary response only; sustained rebalancing after a large disturbance is provided by secondary control, under-frequency load shedding, and hourly dispatch, which are outside the screen. Parameter values and citations are listed in the Supplementary.

3. Case study and assumptions

The case study is a representative small Philippine off-grid island with a peak demand of 1.65 MW—the scale of operational island microgrids such as the Sabang (Palawan) PV–battery–diesel system and typical of the country's several hundred NPC-SPUG off-grid islands [3, 4]. The solar resource is real measurement-derived data: NASA POWER hourly all-sky surface shortwave downward irradiance (CERES SYN1deg) and 2-m air temperature (MERRA-2) for calendar year 2025 at the grid cell centred on 11.0°N, 123.0°E in the central Visayas. The PV capacity factor is obtained from the measured irradiance G_t (W m^{-2}) and air temperature T_t^{air} via a nominal-operating-cell-temperature (NOCT) model,

$$T_t^{\mathrm{cell}} = T_t^{\mathrm{air}} + 0.0256\,G_t,$$
$$\gamma_t = \mathrm{clip}\Big[\tfrac{G_t}{1000}\,\eta_{\mathrm{d}}\,\big(1+\beta\,(T_t^{\mathrm{cell}}-25)\big),\,0,\,1\Big], \tag{10}$$

with system derate $\eta_{\mathrm{d}} = 0.82$, temperature coefficient $\beta = -0.004\,^{\circ}\mathrm{C}^{-1}$, and the cell-temperature coefficient 0.0256 corresponding to NOCT $\approx 40.5\,^{\circ}\mathrm{C}$; this yields an annual mean capacity factor of 16.2%, consistent with fixed-tilt PV at this tropical latitude. The hourly demand is a representative load profile calibrated to a 1.65 MW peak and a 0.72 load factor, with morning, evening, and commercial peaks, weekend reduction, and tropical-tourism seasonality; the weather is real while the load is a representative archetype, an approach standard for islands lacking public hourly metered demand.

Base global parameters are $i = 0.07$, $L = 20$ yr (CRF = 0.0944), $\varepsilon^{\mathrm{dg}} = 0.70$ t $\mathrm{CO_2}$ $\mathrm{MWh^{-1}}$, $\alpha = 0.55$, $\eta^{\mathrm{el}} = 0.67$, $\eta^{\mathrm{fc}} = 0.50$, $\eta^{\mathrm{ch}} = \eta^{\mathrm{dis}} = 0.95$, $c^{\mathrm{pen}} = 10{,}000$ and VOLL = 5000 USD $\mathrm{MWh^{-1}}$. The base carbon price is $c^{\mathrm{co_2}} = 0$; the threshold analysis sweeps it with 150 USD $\mathrm{t^{-1}}$ as the central value. Because only the diesel portfolio emits, the carbon price affects the diesel-battery cost (and the crossover) but leaves the other two unchanged. Unit costs are in Table 1. After the hydrogen-tank right-sizing described in Section 4.1, the portfolios are sized as: battery-only (PV 7 MW / battery 2.2 MW / 24 MWh), diesel-battery (PV 2.8 MW / 1.5 MW / 8 MWh / diesel 2.4 MW), and battery-hydrogen (PV 14 MW / 2.0 MW / 12 MWh / electrolyser 5 MW / fuel cell 2.2 MW / tank 10,000 kg).

Table 1: Techno-economic parameters (from the released configuration).

Component / parameter	Value	Notes
PV capital	1200 USD/kW	
Battery power capital	300 USD/kW	
Battery energy capital	250 USD/kWh	
Diesel genset capital	450 USD/kW	
Electrolyser capital	900 USD/kW	
Fuel-cell capital	1400 USD/kW	
Hydrogen-tank capital	500 USD/kg	
Fixed O&M rate ϕ	0.025 of CAPEX/yr	flat fraction of total CAPEX
Battery throughput c^{thr}	1.0 USD/MWh	on charge + discharge
Electrolyser variable c^{el}	2.0 USD/MWh	
Fuel-cell variable c^{fc}	4.0 USD/MWh	
Diesel fuel (base) c^{fuel}	220 USD/MWh	delivered; swept in Results
Carbon price (base) $c^{\mathrm{co_2}}$	0 USD/tCO$_2$	central sweep value 150
Diesel emission factor $\varepsilon^{\mathrm{dg}}$	0.70 tCO$_2$/MWh	
Battery charge/discharge eff.	0.95 / 0.95	
Electrolyser / fuel-cell eff. (LHV)	0.67 / 0.50	round-trip 33.5%
Hydrogen LHV	33.33 kWh/kg	
Initial SOC / H$_2$ inventory	0.5 of capacity	cyclic to same
Unserved penalty c^{pen}	10,000 USD/MWh	LP device
Value of lost load (resilience)	5000 USD/MWh	
Discount rate / life	0.07 / 20 yr	CRF = 0.0944

4. Results

4.1. Base portfolios and hydrogen-tank right-sizing

At the unit costs of Table 1 and a zero carbon price, the three portfolios have the annual costs and levelised costs of Table 2. The battery-only portfolio is dominated by unserved-energy penalty—PV and a diurnal battery alone cannot firm the load through extended low-irradiance periods—and is retained only as a renewable-only reference; the analysis that follows compares diesel-battery and battery-hydrogen.

A pre-analysis of the hydrogen design exposed a capital oversizing carried by the original fixed-capacity specification. The 50,000 kg hydrogen tank accounted for 47% of the battery-hydrogen capital while providing storage an order of magnitude beyond the design's actual need, which is bounded by two requirements that are both far below 50,000 kg. First, the cost-optimal annual dispatch never stores more than 5,595 kg: the hydrogen inventory follows a seasonal trajectory, drawn down to empty in mid-January and refilled to its 5,595 kg annual peak by December (Fig. 1), so any capacity above this is operationally idle. Second, critical-load resilience is invariant to tank capacity over the tested range 5,000–50,000 kg—the worst-case CLSR remains 1.0 with full 48-h survivability across 50 outage seeds and three durations. The resilience requirement is modest because the binding worst-case outage is a low-irradiance, nighttime-onset event in which the fuel cell must supply the critical load through the dark hours, drawing on the order of 1,411 kg of hydrogen over 48 h—well within the 5,000 kg inventory held at 50% of a 10,000 kg tank; favourable daytime outages are instead covered directly by PV and battery, with surplus PV even replenishing the store. We therefore right-size the tank to 10,000 kg, which accommodates the 5,595 kg operational peak with margin and covers the worst-case outage draw several-fold over. This lowers the battery-hydrogen levelised cost from 610 to 381 USD MWh^{-1} with no change in resilience and no change in the optimised operating cost—a pure elimination of capital embedded by the fixed-capacity design. All subsequent results use the 10,000 kg tank; Fig. 1 shows a representative week of dispatch for both portfolios.

Table 2: Base-case annual cost and levelised cost (carbon price 0).

Portfolio	Annual cost (MUSD/yr)	LCOE (USD/MWh)
diesel-battery	2.25	216
battery-hydrogen	3.96	381
battery-only	19.24	2221

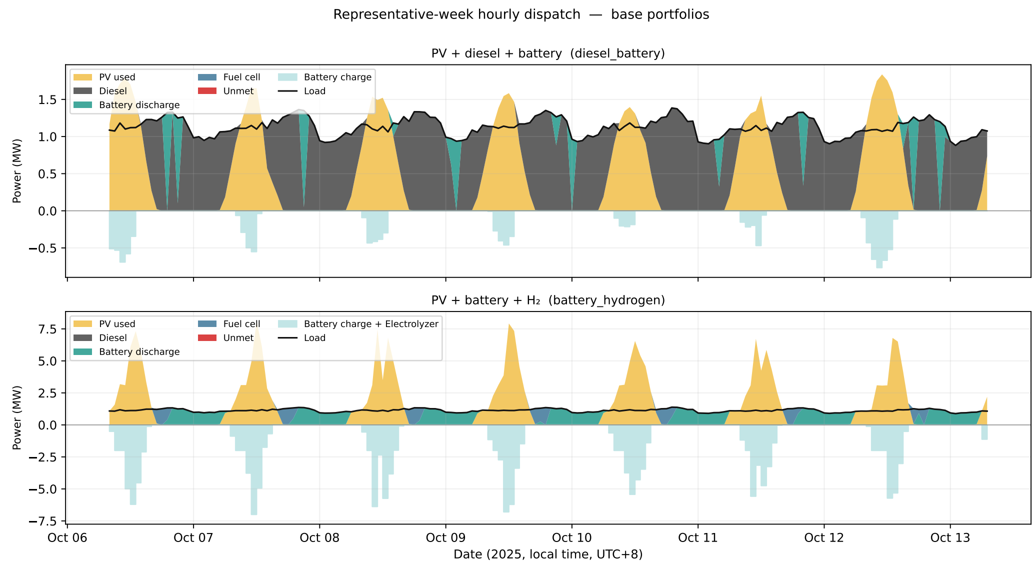

Figure 1: Representative-week dispatch (week of 2025-10-06, in local time UTC+8) for diesel-battery (top) and battery-hydrogen (bottom): daytime PV charges the battery and electrolyser; the battery, and in low-PV periods the fuel cell, serve the evening peak.

4.2. Cost–resilience threshold

Figure 2 maps the annual-cost difference between the two portfolios across delivered diesel cost (100–1000 USD MWh^{-1}) and a hydrogen-CAPEX multiplier (0.30–1.20), at the central carbon price of 150 USD tCO_2^{-1} and a 48-h outage. The zero-difference contour—the economic crossover—passes through **378 USD $\mathbf{MWh^{-1}}$** delivered diesel at the baseline hydrogen CAPEX: below this the diesel-battery portfolio is cheaper, above it battery-hydrogen is. This corresponds to roughly 3.8 USD L^{-1}, within the range reported for remote Philippine islands where maritime fuel logistics dominate delivered cost. Reducing the hydrogen CAPEX to 0.30 of baseline lowers the crossover to 216 USD MWh^{-1}; raising it to 1.20 lifts it to 425 USD MWh^{-1} (Table 3). Internalising the resilience value of avoided unserved energy at VOLL = 5000 USD MWh^{-1} lowers the crossover to **344 USD $\mathbf{MWh^{-1}}$**, because the diesel portfolio's residual outage exposure is then priced. The crossover is interior to the swept diesel range for every multiplier level and reproduces, at the overlapping points, the coarse three-value sensitivity of the original manuscript—now resolved as a continuous frontier.

Table 3: Economic and resilience-adjusted crossover diesel cost versus the hydrogen-CAPEX multiplier μ.

μ (H_2 CAPEX)	Crossover, economic (USD/MWh)	Resilience-adjusted (USD/MWh)
0.30	216	182
1.00	378	344
1.20	425	390

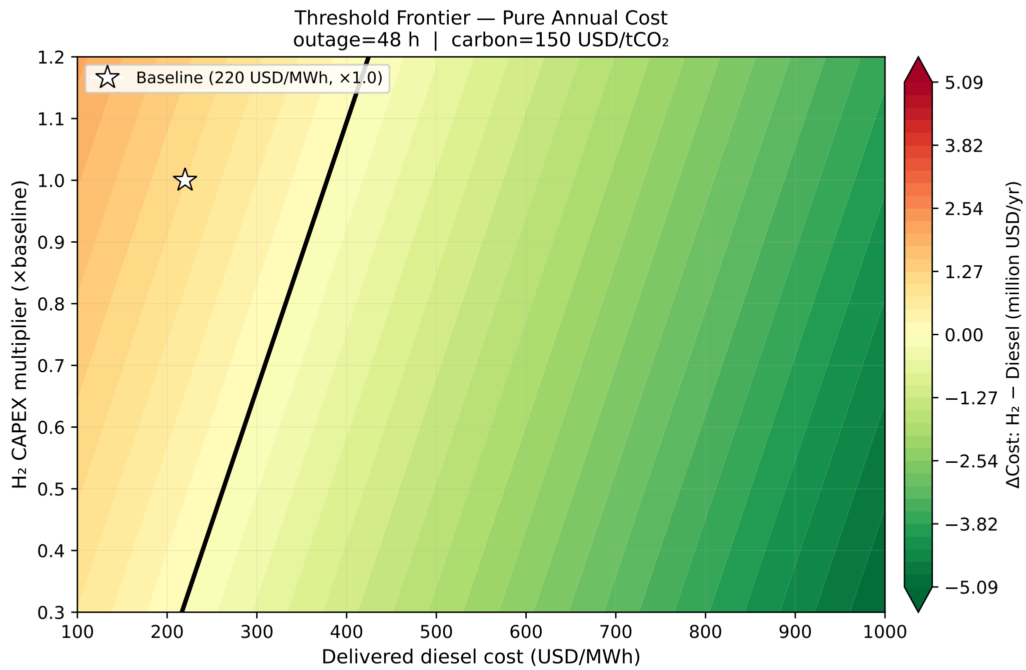

Figure 2: Cost–resilience threshold: annual-cost difference $AC_{BH} - AC_{DB}$ over delivered diesel cost and hydrogen-CAPEX multiplier, at carbon 150 USD tCO_2^{-1} and a 48-h outage. The zero contour is the economic crossover (378 USD MWh^{-1} at $\mu = 1$). The resilience-adjusted contour (VOLL 5000 USD MWh^{-1}) is shown in the Supplementary (Fig. S-fig03b).

4.3. What moves the frontier

Figure 3 decomposes the crossover diesel cost into its sensitivity to each design parameter, varied across its plausible range with the others held at baseline (Table 4). The crossover is most sensitive by far to PV capital cost (span 148 USD MWh^{-1} across 0.6–1.2× of baseline), followed by hydrogen-tank cost (92), electrolyser cost (66), and fuel-cell cost (45); battery-energy cost contributes 11, and the electrolyser and fuel-cell efficiencies move the crossover by less than 1 USD MWh^{-1}. The dominance of PV cost is structural: the battery-hydrogen portfolio carries 14 MW of PV against the diesel-battery portfolio's 2.8 MW, so a change in PV cost bears roughly five-fold more strongly on the hydrogen side; normalised per unit fractional cost change, PV is about 2.5× more influential than the hydrogen tank. The near-zero efficiency sensitivities confirm that the hydrogen subsystem is sized by energy balance and resilience, not by conversion efficiency. The actionable implication is that PV-cost reduction—rather than electrolyser or fuel-cell improvement—is the single most effective lever for hydrogen competitiveness.

Table 4: Cost-driver sensitivity: span of the crossover diesel cost as each parameter is varied across its plausible range.

Parameter (varied 0.6–1.2× baseline unless noted)	Crossover span (USD/MWh)
PV capital cost	148.2
Hydrogen-tank cost	91.9
Electrolyser capital cost	66.2
Fuel-cell capital cost	45.3
Battery-energy cost	11.0
Fuel-cell efficiency	0.7
Electrolyser efficiency	0.6

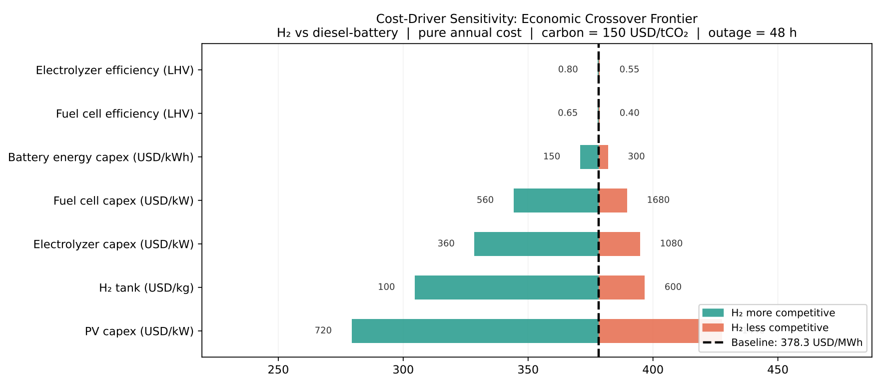

Figure 3: Tornado of the crossover diesel cost: PV capital cost dominates, while converter efficiencies are negligible.

4.4. Is the resilience advantage intrinsic? A matched-backbone test

Section 4.3 shows the frontier is dominated by PV cost, raising the question of whether the hydrogen portfolio's advantages merely reflect its larger PV–battery backbone. Figure 4 isolates the flexibility technology by placing both portfolios on a common backbone—varying PV (7/10/14 MW) and battery energy (8/12 MWh) identically (energy-to-power ratio 6 h) and retaining only the distinguishing diesel or hydrogen subsystem (Table 5). Two results follow. First, the hydrogen portfolio requires a PV backbone of at least 14 MW to be viable at all: at 7–10 MW the electrolyser cannot be supplied enough surplus to balance the 33.5%-efficient hydrogen round-trip, the portfolio incurs large unserved-energy penalties, and no crossover exists within the swept diesel range. The large PV of the hydrogen design is thus a physical requirement, not arbitrary oversizing. Second, and decisively, the resilience gap persists under a matched backbone: at an identical 14 MW / 12 MWh backbone the diesel-battery portfolio sheds critical load in the worst-case 48-h outage—its worst-case survivable duration is zero hours and its worst-case CLSR is 0.98 across ten seeds—while battery-hydrogen sustains the full 48 hours (CLSR 1.0), and the diesel-battery worst-case survivability is zero at every backbone tested whereas battery-hydrogen reaches full 48-h survivability once the PV backbone is large enough to keep its store charged ($\geq$10 MW with the larger battery, and at 14 MW). Because the only difference at a matched backbone is the flexibility subsystem, the resilience advantage is attributable to the long-duration hydrogen store itself, not to fixed capacities.

Two points follow for interpretation. The matched-backbone economic crossover is not the headline threshold: forcing the diesel-battery portfolio onto hydrogen's 14 MW PV spine is not its cost-optimal configuration, so the matched crossover (368 USD MWh^{-1} at 14/8 MWh, 755 at 14/12 MWh) brackets rather than replaces the 378 USD MWh^{-1} obtained when each technology is compared at its own appropriate scale. The matched backbone is therefore used to isolate the *resilience* contribution of the flexibility technology, not to redefine the economic threshold; on the economic axis it confirms only that hydrogen's competitiveness is sensitive to how much PV the comparison grants the diesel alternative.

Table 5: Matched-backbone comparison. Crossover "—" denotes none within the swept diesel range (the hydrogen portfolio incurs large annual unserved-energy penalties below 14 MW PV). $CLSR_{min}$ and survivable hours are worst-case over ten seeds for a 48-h outage with diesel unavailable.

Backbone (PV / battery)	Crossover (USD/MWh)	Diesel–battery $CLSR_{min}$	Diesel–battery surv. (h)	Battery–H_2 $CLSR_{min}$	Battery–H_2 surv. (h)
7 MW / 8 MWh	—	0.88	0	0.98	9
10 MW / 12 MWh	—	0.96	0	1.00	48
14 MW / 8 MWh	368	0.92	0	1.00	48
14 MW / 12 MWh	755	0.98	0	1.00	48

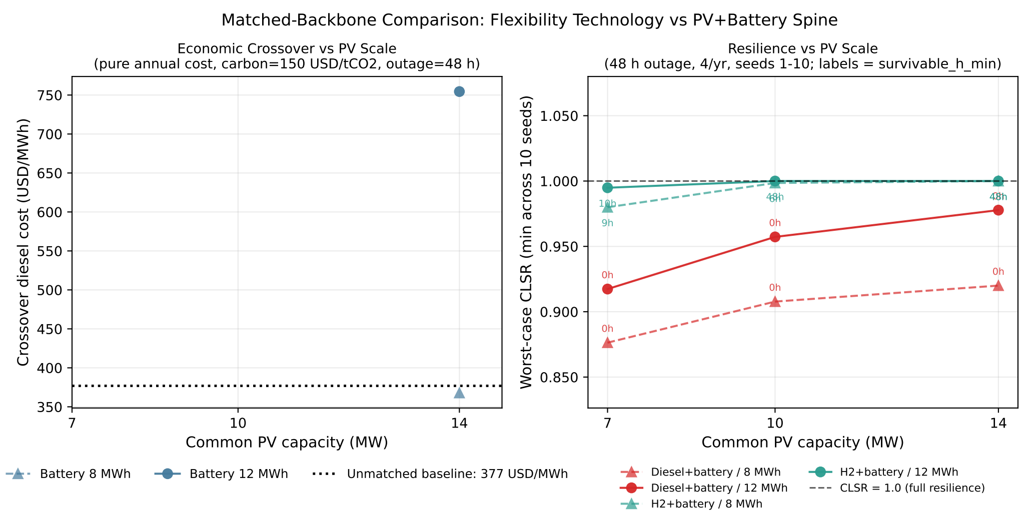

Figure 4: Matched-backbone comparison across common PV and battery sizes: battery-hydrogen requires ≥14 MW PV to be feasible, and at a matched backbone retains full 48-h survivability while diesel-battery sheds critical load (worst-case survivable duration 0 h).

4.5. *Robustness to the diesel-support assumption*

The resilience comparison so far assumes diesel is unavailable during an outage. Figure 5 relaxes this across graded support cases. Under full unavailability (D0) the diesel-battery worst-case survivability is zero hours and the worst-case CLSR is 0.58 across ten seeds—the single-seed value of 0.65 in the original manuscript was optimistic. Restoring support closes the gap only above clear thresholds: the diesel-battery portfolio matches the hydrogen portfolio's 48-h survivability only with $\geq$50% derating (D1) or $\geq$12 h of fuel (D2). Most strikingly, any start-up or delivery delay collapses worst-case survivability to zero—even a six-hour delay (D3)—because the battery depletes before diesel arrives during the worst-timed outage. With diesel fully available but its delivered fuel cost shocked (D4), the economic crossover is unchanged at 378 USD MWh^{-1}; above roughly 380 USD MWh^{-1} the battery-hydrogen portfolio is simultaneously cheaper and at least as resilient. The hydrogen portfolio's value is therefore specifically a hedge against degradation of diesel support—derating, fuel limits, or delivery delay—the characteristic failure modes during storms and logistics disruptions.

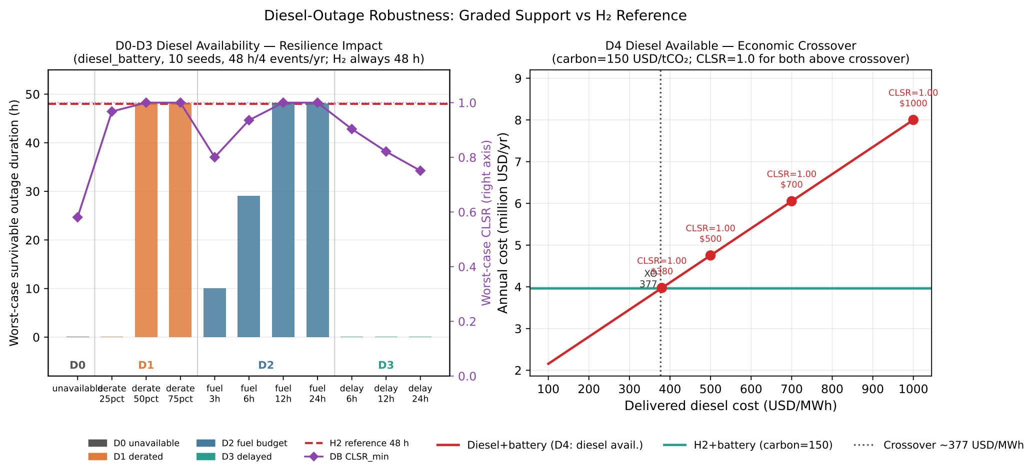

Figure 5: Graded diesel-support robustness. Worst-case survivability of diesel-battery reaches the hydrogen reference (48 h) only with $\geq$50% derating (D1) or $\geq$12 h fuel (D2); any delay (D3) returns it to zero. battery-hydrogen is the invariant reference.

4.6. Dynamic feasibility

Figure 6 screens primary-frequency response (Table 6). Under a 10% load step all three portfolios hold the frequency nadir above 49.0 Hz; the RoCoF is 0.86 Hz s^{-1} for diesel-battery (within both islanded and mainland limits, owing to synchronous-generator inertia), 1.88 Hz s^{-1} for battery-only, and 2.06 Hz s^{-1} for battery-hydrogen—the last marginally above the 2.0 Hz s^{-1} islanded screening value, which a modest increase of the grid-forming virtual-inertia constant to $\geq$ 1.2 s removes. The portfolios diverge sharply only in the contingency that matters most: loss of the diesel generator. When the

1.65 MW diesel trips and the 1.5 MW battery must assume grid-forming, the instantaneous RoCoF reaches 27 Hz s^{-1} (5.9 Hz s^{-1} over the relay-relevant 500 ms window) and the nadir falls to 45.9 Hz, because the battery is undersized relative to the generation it must instantly replace; full rebalancing requires under-frequency load shedding, though the battery alone sustains the 0.55 critical fraction (1.5 MW $\geq$ 0.91 MW). The diesel-battery portfolio is thus dynamically weakest during exactly the event—diesel loss—at which Sections 4.4–4.5 found it energetically weakest, whereas battery-hydrogen, with a larger grid-forming battery, fuel-cell droop support, and no diesel to lose, avoids the contingency. The dynamic screen reinforces rather than qualifies the cost–resilience conclusion.

Table 6: Primary-frequency screening metrics. Screening limits: nadir $\geq$ 49.0 Hz, $|\text{RoCoF}| < 2.0$ Hz s^{-1} (islanded).

Disturbance	Portfolio	Nadir (Hz)	RoCoF (Hz/s)	Note
10% load step	diesel-battery	>49	0.86	pass
10% load step	battery-only	>49	1.88	pass
10% load step	battery-hydrogen	>49	2.06	marginal (VSM tuning)
Diesel trip	diesel-battery	45.9	27.5	5.9 Hz/s at 500 ms; UFLS, critical load held

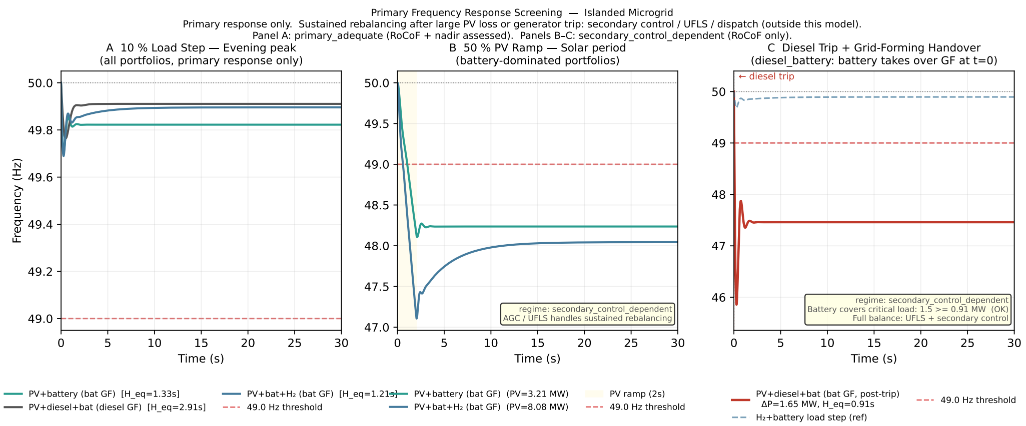

Figure 6: Reduced-order primary-frequency response. Load step (left): all portfolios hold nadir >49 Hz. Diesel trip (right): diesel-battery exhibits a deep nadir and high RoCoF during the grid-forming hand-over.

4.7. Synthesis

Taken together (Figs. 2–6), the battery–hydrogen portfolio is the defensible design under either of two independently realistic conditions: a delivered diesel cost at or above 378 USD MWh^{-1} (Sections 4.2, 4.5), or any material degradation of diesel support during outages (Section 4.5), under which the diesel–battery worst-case survivability collapses—in both energy (Sections 4.4–4.5) and frequency-dynamic (Section 4.6) terms—while the hydrogen store holds full critical-load service. Both conditions characterise remote-island and typhoon contingencies. The hydrogen design's large PV is a physical requirement of the round-trip (Section 4.4), and PV-cost reduction is the most powerful lever on its competitiveness (Section 4.3).

5. Discussion

5.1. Principal findings

The analysis yields a single, design-actionable picture rather than the qualitative claim that hydrogen becomes attractive when diesel is expensive. The economic crossover between the two portfolios is a continuous frontier whose central value is 378 USD MWh^{-1} of delivered diesel at baseline hydrogen cost and a 150 USD tCO_2^{-1} carbon price, falling to 344 USD MWh^{-1} once the resilience value of avoided unserved energy is internalised, and ranging from 216 to 425 USD MWh^{-1} as hydrogen capital varies over 0.30–1.20×.

The position of that frontier is governed overwhelmingly by PV capital cost—2.5 times more strongly than by hydrogen-tank cost and two orders of magnitude more strongly than by converter efficiency—because the hydrogen portfolio is intrinsically PV-intensive (14 versus 2.8 MW) to balance a 33.5%-efficient round-trip. The resilience advantage is not an artefact of the larger backbone: on an identical PV–battery backbone the diesel–battery portfolio sheds critical load in a worst-case 48-h outage (zero guaranteed survivable hours, worst-case CLSR 0.98) while the hydrogen portfolio sustains the full duration, so the advantage is attributable to the long-duration store itself. That advantage is conditional on the diesel-unavailability assumption; relaxing it shows the diesel portfolio matches hydrogen only above clear support thresholds ($\geq$50% derate or $\geq$12 h fuel) and that any start-up delay—even six hours—collapses its worst-case survivability. Finally, the diesel portfolio is dynamically weakest at precisely the diesel-trip contingency where it is energetically weakest, with a 500 ms RoCoF of 5.9 Hz s^{-1} and a 45.9 Hz nadir requiring load shedding, whereas the hydrogen portfolio has no equivalent single-point contingency.

5.2. Why the result is not self-evident

That hydrogen wins when diesel is dear is, taken alone, unsurprising. The contribution here is the structure behind that statement, which is neither obvious a priori nor recoverable from a coarse break-even calculation. First, the crossover is set not by the hydrogen chain's own cost or efficiency but by PV capital cost—a consequence of the portfolio's energy balance that only a full 8760-h optimisation reveals, and one that redirects cost-reduction effort from converters toward PV. Second, the resilience gap survives a matched backbone: a reader could reasonably attribute hydrogen's outage performance to its larger PV and battery, and the matched-backbone experiment is required to show that it does not. Third, the diesel portfolio's resilience is far more fragile than a single-seed or perfect-availability analysis implies; the finding that a six-hour diesel-start delay erases worst-case survivability entirely, while a 50% derate or a 12-h fuel limit does not, is a non-monotone, operationally specific result. Fourth, the most dangerous frequency event for the diesel portfolio is the loss of the very asset that provides its inertia, so that the technology improving small-signal RoCoF also creates the binding large-disturbance contingency. Each is a design-actionable, non-obvious refinement of the headline.

5.3. Relation to prior work

The individual techniques used here are established; the contribution is their integration and the attribution it enables. Prior island studies optimise component sizing with tools such as HOMER and mixed-integer programs [9, 10], and recent work has tested fuel-constrained or graded-availability outages [15, 25] and combined techno-economic with dynamic assessment [23, 24]. What has been missing is a single framework that (i) resolves the cost–resilience trade-off as a continuous, carbon-priced threshold rather than a point estimate, (ii) attributes that threshold to specific design parameters, (iii) isolates the contribution of the flexibility technology from that of the shared backbone, and (iv) verifies dynamic feasibility for the same portfolios and contingencies. The matched-backbone experiment (Section 4.4) is the methodological centrepiece: by holding the PV–battery spine fixed it separates the effect of the long-duration store from that of capacity, which sizing-only optimisations cannot do. Relative to survivability-focused resilience studies [15, 16, 28], the present work links the survivability metric to the economic threshold and to a primary-frequency screen on the same system; relative to integrated techno-economic–dynamic studies, it adds the attribution and technology-isolation analyses and grounds the comparison in a real measured-weather year for a representative Philippine off-grid context [4, 11]. The Philippine setting is not incidental: with several hundred NPC-SPUG diesel islands and operational PV–battery–diesel microgrids, delivered diesel costs and outage exposure of the magnitude examined here are realistic rather than hypothetical [3, 29].

5.4. Limitations and future work

Several limitations bound the claims and define the next steps. The dispatch uses a single measured weather year (2025) at one grid cell; interannual variability and a typhoon-year stress test are not captured and would refine both the cost and the resilience estimates. The demand is a representative calibrated archetype rather than metered island load, because public hourly demand for individual off-grid islands is scarce; substituting a metered profile would sharpen the demand-side detail without altering the structure of the comparison. The operational model is a deterministic linear program with perfect foresight, which yields a lower bound on operating cost; rolling-horizon or stochastic operation would test the robustness of the cost figures, although the resilience results derive from a separate, foresight-free outage simulation. The frequency analysis is a reduced-order primary-response screen; it establishes feasibility and ranks contingencies but is not a substitute for electromagnetic-transient simulation, secondary/automatic generation control, under-frequency load-shedding design, or protection coordination, which are needed before deployment and which the marginal battery-hydrogen RoCoF and the diesel-trip nadir both motivate. The system is a single island with PV as the only renewable resource; wind hybridisation, inter-island interconnection, and component degradation are left to future work. Finally, hydrogen costs are evolving rapidly; the capital-multiplier sweep partially internalises this, but the crossover should be re-evaluated as cost data mature.

Two further caveats sharpen these points. First, outage start times are sampled uniformly across the year rather than correlated with the adverse-weather episodes—multi-day typhoon overcast—that most stress an islanded system; although the worst-case-over-seeds statistic captures low-irradiance onsets, explicitly conditioning outages on a typhoon-affected weather year would be a more demanding and realistic resilience test. Second, the flexibility components (electrolyser, fuel cell, tank, and diesel rating) are held at representative fixed sizes; co-optimising these alongside the PV–battery backbone would refine the absolute thresholds, though the matched-backbone design already controls for the dominant PV–battery sizing effect.

6. Conclusion

For islanded power systems, the choice between a PV–diesel–battery and a PV–battery–hydrogen portfolio is governed by a quantifiable cost–resilience threshold rather than a qualitative preference. Using an 8760-h cost-optimal dispatch model driven by real NASA POWER weather for a representative Philippine off-grid island, together with a worst-case outage simulation and a reduced-order frequency screen, we find that the battery–hydrogen portfolio is the defensible design under either of two independently realistic conditions: a delivered diesel cost at or above 378 USD MWh^{-1} (344 USD MWh^{-1} when avoided unserved energy is valued), or any material degradation of diesel support during outages—a derate below 50%, a fuel budget under 12 hours, or any start-up delay—each of which collapses the diesel portfolio's worst-case survivability to near zero in both energy and frequency-dynamic terms while the hydrogen store sustains full critical-load service for 48 hours. The threshold's position is set principally by PV capital cost, making PV-cost reduction the most effective lever on hydrogen competitiveness, while converter efficiency is nearly immaterial; and a matched-backbone test shows the resilience advantage to be intrinsic to the long-duration hydrogen store rather than a by-product of the larger backbone the store requires. These conditions correspond to exactly the remote-island and storm contingencies that motivate off-grid hydrogen, converting a qualitative intuition into design-actionable thresholds, attributions, and feasibility checks that island planners can apply directly.

Data availability

All code, model configuration, input data (real NASA POWER 2025 weather and the calibrated load profile), and the scripts that generate every figure and table are openly available at https://github.com/exoprime-sketch/microgrid-h2-paper and archived at Zenodo (DOI 10.5281/zenodo.19658748).

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the author used AI-assisted tools for language editing.

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